

# TADP-Sec Phase-1 Control Scope Matrix

Implemented Phase-1 controls and explicitly excluded Phase-2 / adversarial-ML claims

| Control or threat                        | T1                  | T2                | T3                               | T4                | Phase-1 interpretation                                                                                               |
|------------------------------------------|---------------------|-------------------|----------------------------------|-------------------|----------------------------------------------------------------------------------------------------------------------|
| Invalid signature                        | Reject              | Reject            | Reject                           | Reject            | RSA-PSS verification before aggregation                                                                              |
| Replay / stale submission                | Reject              | Reject            | Reject                           | Reject            | Nonce, round, and freshness validation                                                                               |
| PoFC rate-limit overflow                 | Reject              | Reject            | Reject                           | Reject            | Packet limit; client removal; session reform                                                                         |
| Tier / metadata tampering                | Reject              | Reject            | Reject                           | Reject            | Signed envelope plus server-state validation                                                                         |
| Numerical fidelity / consistency failure | N/A (plain path)    | Abort             | Abort                            | Abort             | Protected paths must pass frozen numerical gates                                                                     |
| Update confidentiality in aggregation    | Signed plain update | DF + KS           | xMK-CKKS                         | SAMK              | Tier-specific aggregation protection                                                                                 |
| Client dropout availability              | No dedicated test   | No dedicated test | All-share xMK may require re-key | Tested separately | T4_TEST_DROPOUT: survivor aggregation after pre-upload dropout + delayed post-upload recovery; locked tier preserved |
| Semantic poisoning / backdoor            | Out of scope        | Out of scope      | Out of scope                     | Out of scope      | Future adversarial-ML evaluation                                                                                     |
| Dynamic likelihood-driven escalation     | Out of scope        | Out of scope      | Out of scope                     | Out of scope      | Future Phase-2 governance layer                                                                                      |